

Object: To determine the value of acceleration due to gravity and radius of gyration using bar pendulum

Apparatus used: Bar pendulum, stop watch and meter scale.

Formula used:

A. The general formula of the time period for bar pendulum is given by following equation:

$$2\pi \sqrt{\frac{\frac{k^2}{l} + l}{g}} = 2\pi \sqrt{\frac{l_2 + l_1}{g}}$$

$$g = \frac{4\pi^2(l_1 + l_2)}{T^2} = \frac{4\pi^2 L}{T^2} \quad \text{--- (1)}$$

Where  $l$  distance between C.G. and suspension point,  $L$ : distance between suspension and oscillation points,

$$L = l_1 + l_2 = l + \frac{k^2}{l}$$

$g$ : acceleration due to gravity,  $T$ : time period

B. The time period is minimum when  $l = \pm k$  in this situation the equation (1) becomes as:

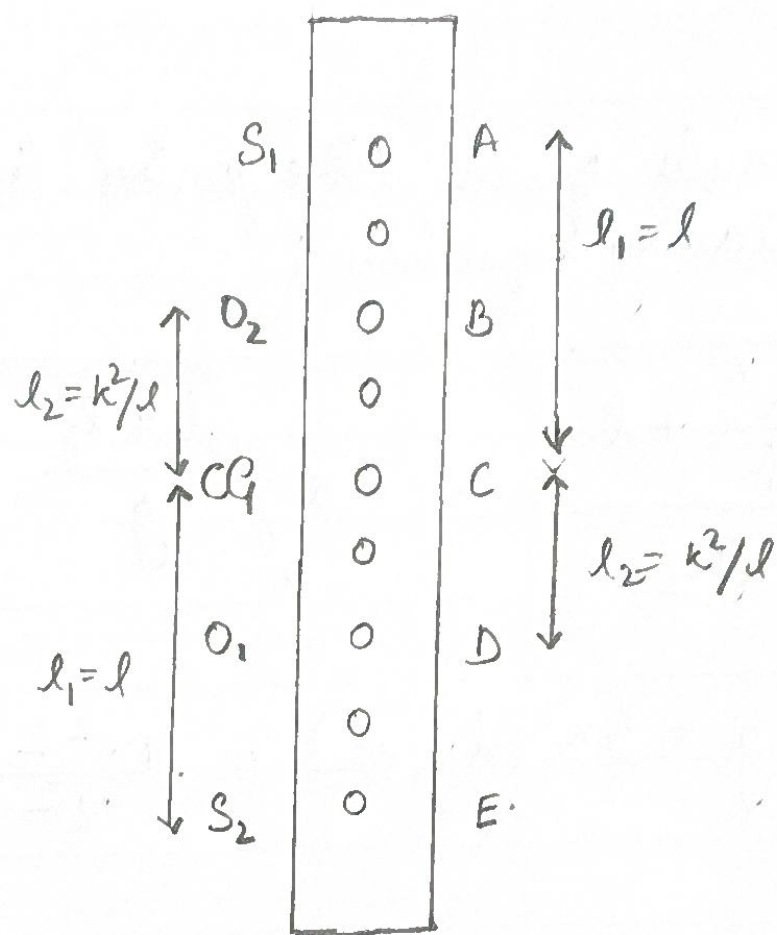


Fig. 1

$$T_{\min} = 2\pi \sqrt{\frac{2k}{g}}$$

$$g = \frac{8\pi^2 k}{T_{\min}^2} \quad \text{--- (II)}$$

Or

where,  $k$ : radius of gyration,  $T_{\min}$ : minimum time period

The value of 'g' can be calculated using equations (I) and (II).

The values of  $L$ ,  $T$ ,  $k$  and  $T_{\min}$  are obtained using graph between  $T$  and  $L$  for bar pendulum which is shown in following figure.

From figure (1) and (2)

- (a)  $L_1 = AC + CD$ ,  $L_2 = EC + CB$  and  $L = (L_1 + L_2)/2$ ,  $T$  = time at C
- (b)  $k = (PQ + QR)/2$  and  $T_{\min}$  = time at P

C. The radius of gyration can be obtained with following formula

$$k = \sqrt{L_1 L_2}$$

Where  $L_1 = (AC + CE)/2$ ,  $L_2 = (BC + CD)/2$

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Calculation

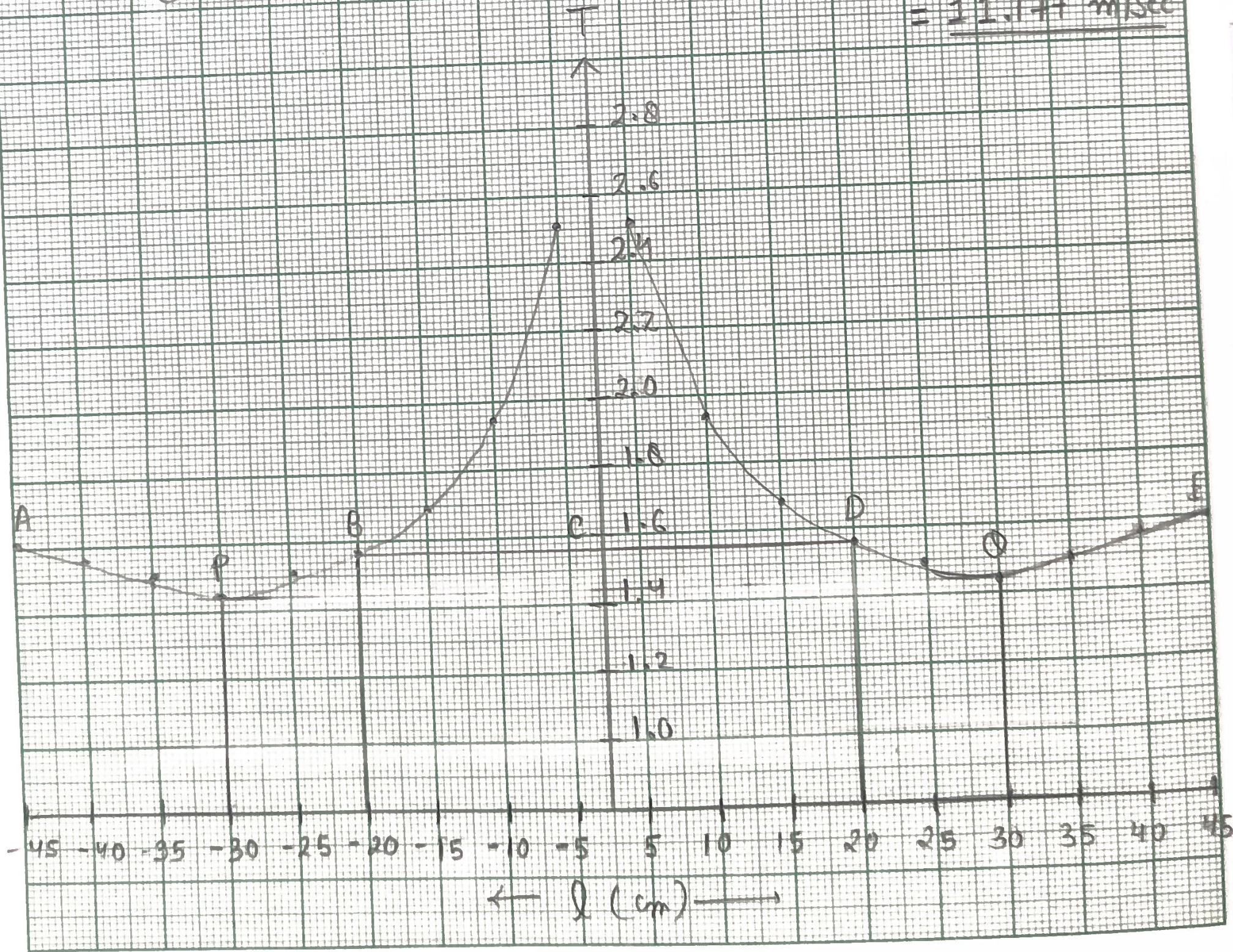
$$l_1 = \frac{AE}{2} = \frac{90}{2} = 45, \quad l_2 = \frac{BD}{2} = \frac{40}{2} = 20$$

$$\Rightarrow K = \sqrt{l_1 l_2} = \sqrt{45 \times 20} = \sqrt{900} = 30 \text{ cm}$$

$$\Rightarrow K = \text{from graph} = \frac{PQ}{2} = \frac{60}{2} = 30 \text{ cm}$$

$$\Rightarrow g = \frac{8\pi^2 K}{(T_{\min})^2} = \frac{8\pi^2 \times 30}{(1.455)^2} = \frac{2366.304}{2.117025} = 1117.74 \text{ cm/sec}^2$$

$$= \underline{\underline{11.177 \text{ m/sec}^2}}$$



Observations :

1. Least count of the stop watch = 0.1 sec
2. Least count of the meter scale = 0.1 cm
3. Table for  $l$  and  $T$

| S.No.                           | $l$ (cm) | $t$ (time taken for 10 oscillation) | $T = t/10$ |
|---------------------------------|----------|-------------------------------------|------------|
| For first half side of the bar  |          |                                     |            |
| 1.                              | 45       | 15.53                               | 1.553      |
| 2.                              | 40       | 15.47                               | 1.547      |
| 3.                              | 35       | 14.95                               | 1.495      |
| 4.                              | 30       | 14.55                               | 1.455      |
| 5.                              | 25       | 14.96                               | 1.496      |
| 6.                              | 20       | 15.60                               | 1.560      |
| 7.                              | 15       | 16.66                               | 1.666      |
| 8.                              | 10       | 18.70                               | 1.870      |
| 9.                              | 5        | 25.85                               | 2.585      |
| For second half side of the bar |          |                                     |            |
| 10.                             | -5       | 25.85                               | 2.585      |
| 11.                             | -10      | 18.70                               | 1.870      |
| 12.                             | -15      | 16.66                               | 1.666      |
| 13.                             | -20      | 15.60                               | 1.560      |
| 14.                             | -25      | 14.96                               | 1.496      |
| 15.                             | -30      | 14.55                               | 1.455      |
| 16.                             | -35      | 14.95                               | 1.495      |
| 17.                             | -40      | 15.47                               | 1.547      |
| 18.                             | -45      | 15.53                               | 1.553      |

Teacher's Signature \_\_\_\_\_

Results: The acceleration due to gravity ( $g$ ) = \_\_\_\_\_  $m/s^2$   
 Radius of gyration ( $k$ ) = \_\_\_\_\_ cm (from calculation)  
 = \_\_\_\_\_ cm (from graph)

### Precautions:

1. The motion of the pendulum should be in a vertical plane. While taking the time, start taking observations after two oscillations to avoid any irregularity of motion.
2. The amplitude of oscillation should be small.

### Viva-voice Questions:-

Ques1 What is a bar pendulum, and how does it differ from a simple pendulum?

Ans A bar pendulum is a physical apparatus used to study oscillations and determine the period of a pendulum with a different shape, typically a bar or a rod, instead of a simple point mass. Unlike a simple pendulum, which is modelled as a point mass attached to a string or rod, a bar pendulum considers the mass distribution along the length of the bar making it more complex to analyse.

Ques2 Describe the setup of a bar pendulum experiment

Ans In a bar pendulum experiment, you typically have a horizontal bar or rod that is free to rotate about a fixed axis. One end of the bar is released from a horizontal position and it oscillates back and forth. The oscillations are usually measured by timing

how long it takes for the bar to complete a certain number of cycles.

Ques 3 How can you determine the period of a bar pendulum experimentally?

Ans To determine the period of a bar pendulum experimentally you can follow these steps:

1. Measure the length of a bar from the axis of rotation to the centre of mass.
2. Displace the bar from its equilibrium position (usually horizontal) by a small angle.

Ques 4 How does the period of a bar pendulum differs from that of a simple pendulum?

Ans Simple Pendulum: The time period depends only on its effective length ( $T = 2\pi \sqrt{l/g}$ ) because its mass is concentrated as a point bob.

Bar Pendulum: It is a compound pendulum, so its time period depends on both its length and the mass distribution (moment of inertia) relative to the pivot ( $T = 2\pi \sqrt{\frac{k^2 + l^2}{gl}}$ ), where  $k$  is the radius of gyration.

Ques 5 What mathematical equations or principles can be used to analyse the motion of a bar pendulum?

Ans • Principal Equation: The time period is given by :

$$T = 2\pi \sqrt{\frac{I}{mgl}} = 2\pi \sqrt{\frac{k^2 + l^2}{gl}}$$

(where  $I$  is moment of inertia,  $m$  is mass,  $l$  is distance from pivot to center of gravity, and  $k$  is radius of gyration).

• Governing Principle: Torque due to gravity restoring the bar to equilibrium

( $\tau = -mgl \sin \theta$ ). For small angles ( $\sin \theta \approx \theta$ ), the motion is Simple Harmonic Motion (SHM).

Ques 6 How can the period of a bar pendulum be affected by changing the amplitude of oscillation?

Ans • Small Amplitude ( $\theta < 5^\circ$ ): The period remains constant and independent of amplitude because  $\sin \theta \approx \theta$  holds true (isochronous motion).

• Large Amplitudes: The approximation fails, causing the time period to slightly increase as amplitude increases.